

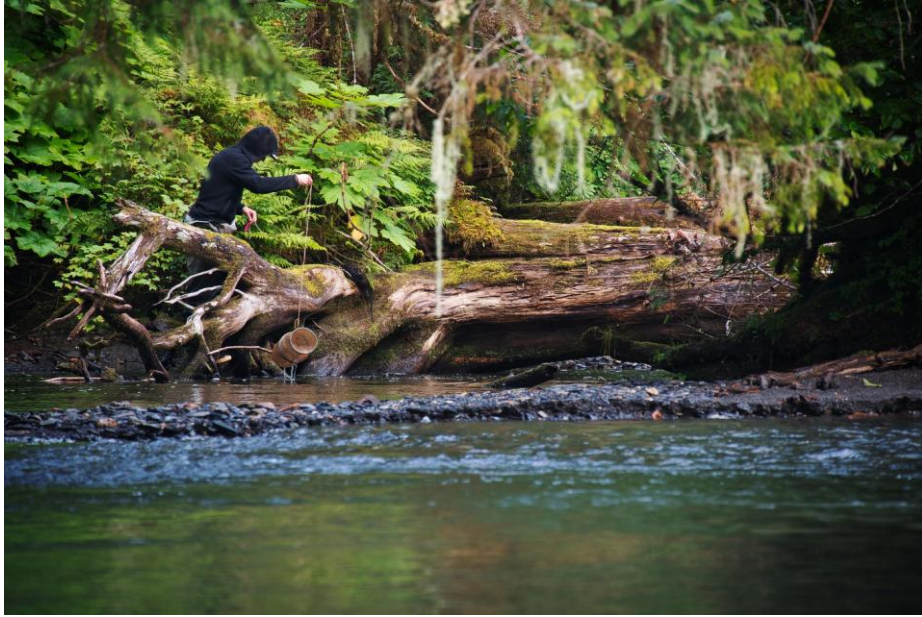
Alaska Cooperative Fish and Wildlife Research Unit

Annual Research Report: 2026



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Setting fish traps in Southeast Alaska.
Photo credit: Blake Toney.

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Not for Publication: Because this report is one of progress, the data presented are often incomplete, and the conclusions reached may not be final. Consequently, permission to publish any of the information herein is withheld pending approval from the Alaska Cooperative Fish and Wildlife Research Unit.

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Unit Roster

Federal Scientists

- Jeff Muehlbauer: Unit Leader–Fisheries
- Megan Milligan: Assistant Unit Leader–Wildlife
- *Vacant*: Assistant Unit Leader (x3)

University Staff

- Monica Armbruster: Fiscal Professional
- Matt Kynoch: Research Associate
- *Vacant*: Administrative Generalist

Unit Students

- Nate Cathcart, PhD Fisheries (Muehlbauer)
- Tim Ericson, MS Fisheries (Muehlbauer)
- Thomas House, MS Fisheries (Muehlbauer)
- Ollie Leibovich, PhD Fisheries (Muehlbauer)
- Kayleigh McCarthy, MS Wildlife Biology and Conservation (Crimmins)
- Riley Marcus, MS Wildlife Biology and Conservation (Milligan)
- Arrow Myers, MS Wildlife Biology and Conservation (Milligan)
- Christie Osburn, MS Wildlife Biology and Conservation (Crimmins, Milligan)
- Hunter Parini, MS Fisheries (Muehlbauer)
- Maddy Rusho, MS Biological Sciences (Muehlbauer)
- Rebecca Shaftel, PhD Fisheries (Falke)
- Sam Simon, MS Wildlife Biology and Conservation (Crimmins)
- Miles Spathelf, PhD Biological Sciences (Crimmins, Milligan)
- Sarah Swanson, PhD Biological Sciences (Crimmins)
- Will Tangredi, MS Biological Sciences (Muehlbauer)
- Sarah Trujillo, PhD Biological Sciences (Crimmins)
- Danielle Tryon, MS Fisheries (Muehlbauer)
- Jacob Tsikoyak, MS Fisheries (Muehlbauer)
- Hannah Vincelette, MS Wildlife Biology and Conservation (Crimmins, Milligan)
- Lauren Yancy, MS Fisheries (Muehlbauer)

Unit Post-Doctoral Researchers

- Derek Arnold (Crimmins)
- Matt Dunkle (Muehlbauer)

Unit Students Graduated This Year

- John Hermus, MS Fisheries (Muehlbauer)
- Ian Kwit, MS Biological Sciences (Muehlbauer)
- Dakota Keller, MS Biological Sciences (Muehlbauer)
- Eli Wilson, MS Fisheries (Muehlbauer)
- Tony Zambito, MS Wildlife Biology and Conservation (Crimmins)

Affiliated Students

- Isaac Nyameke, PhD Fisheries (Westley)
- Joe Spencer, MS Fisheries (Seitz)
- Blake Toney, MS Fisheries (Cunningham)
- Binta Wold, MS Wildlife Biology and Conservation (Brinkman)

Affiliated Students Graduated This Year

- Margaret Harings, MS Fisheries (López and Schoen)
- Elizabeth Lindley, PhD Fisheries (Westley)

University Cooperators

- Peter Bieniek, IARC
- Greg Breed, DBW, IAB
- Todd Brinkman, DBW, IAB
- Ophélie Couriot, DBW, IAB
- Morag Clinton, VetMed, Sitka Sound Science Center
- Curry Cunningham, CFOS
- Arlo Davis, IARC
- Eugénie Euskirchen, DBW, IAB
- Hélène Genet, IAB
- Brad Griffith, IAB
- Nina Hansen, VetMed
- Laura Conner, Vice Chancellor for Research, UAF
- Brian Smentkowski, Vice President for Academics, Students, and Research, UA Statewide
- J. Andrés López, CFOS
- A. David McGuire, IAB
- Charlene Stern, Acting Provost, UAF
- Caleb Robbins, Baylor University
- T. Scott Rupp, IARC, SNAP
- Will Samuel, IARC
- Erik Schoen, IARC
- Tobias Schwoerer, IARC
- Andy Seitz, CFOS
- Michael Sfraga, Acting Chancellor, UAF
- Deanna Strohm, CFOS
- Trent Sutton, CFOS
- Peter Westley, CFOS

Unit Cooperators

- Diane O'Brien—Director, Institute of Arctic Biology, University of Alaska Fairbanks
- Doug Vincent-Lang—Commissioner, Alaska Department of Fish and Game
- Sara Boario—Director, Region 7, U.S. Fish and Wildlife Service
- Jen Newmark—Western Conservation Coordinator, Wildlife Management Institute
- Kevin Whalen—Unit Supervisor, Cooperative Research Units, U.S. Geological Survey

Background

This is the Annual Report for the Alaska Cooperative Fish and Wildlife Research Unit, highlighting activities for State of Alaska fiscal year 2026. The Unit engages in research on living natural resources for a variety of State and Federal agencies. As an unbiased research organization, the Unit provides information requested and funded by these agencies. When studies are completed, the agencies use the information to assist in their natural resource management efforts. Most of the research is conducted by graduate students, many of whom go on to work for the agencies upon graduation.

The Alaska Unit was established in 1950 for wildlife, with the addition of a separate fisheries unit in 1978, and a merger into today's Alaska Cooperative Fish and Wildlife Research Unit in 1991. Across this history, the Unit has provided three-quarters of a century of research dedicated to helping conserve and enhance the living natural resources of the State and the Arctic Region. The Unit is part of a larger and even older program, the U.S. Department of the Interior's Cooperative Research Unit Program. Established in 1935, Cooperative Research Units were created to fill the vacuum of wildlife management information and the shortage of trained wildlife biologists. In 1960, the Unit Program was formally sanctioned by Congress with the enactment of the Cooperative Units Act. Each unit is a partnership between the Ecosystems Mission Area of the U.S. Geological Survey, a host university, a state fish and game agency, the U.S. Fish and Wildlife Service, and the Wildlife Management Institute. Staffed by Federal scientists, Cooperative Research Units conduct research on renewable natural resource questions, participate in the education of graduate students destined to become natural resource managers and scientists, provide technical assistance and consultation to parties who have legitimate interests in natural resource issues, and provide continuing education for natural resource professionals. Presently, there are 43 Cooperative Research Units in 41 states conducting research on virtually every type of North American ecological community. The Program is staffed by more than 100 PhD scientists who advise as many as 675 graduate student researchers per year.

Land Acknowledgement

We acknowledge the Alaska Native nations upon whose ancestral lands our campus resides. The University of Alaska Fairbanks' Troth Yeddha' Campus, where the Unit is located, is on the traditional lands of the Dena people of the lower Tanana River. Our research throughout Alaska takes place on the unceded lands of the Haida, Tsimshian, ƛingít/Tlingit, Eyak, Athabaskan, Yup'ik, Cup'iq, Sugpiaq/Alutiiq, Unanga/Aleut, and Iñupiaq peoples. We respect and appreciate their past and present stewardship of these lands and waters.

Statement of Direction

The research program of the Unit will be aimed at understanding the ecology of Alaska's fish and wildlife, evaluating impacts of land use and development on these resources, and relating effects of social and economic needs to production and harvest of natural populations.

In addition to the expected Unit functions of graduate student training/instruction and technical assistance, research efforts will be directed at problems of productivity, socioeconomic impacts, and perturbation on fish and wildlife populations, their habitats, and ecosystems. Fisheries research will emphasize water quality, habitat characteristics, and life history requirements of northern fish populations. Wildlife research will focus on the ecology of northern birds and mammals and their habitats. Unit research will also be directed at integrated studies of fish and wildlife at the ecosystem level.

Message from the Unit Leader

Dr. Jeff Muehlbauer

This has been a year of adjustment and charting new courses for AKCFWRU. We have navigated new uncertainties to unit operations, accentuated by the extended federal government shutdown in fall 2025. Coming out of this, I have been proud of our unit's adaptability and productivity in seeking out grant funding, matriculating new students, and meeting the needs of our state, federal, and university cooperators. Those efforts shine throughout this report.

At the unit level, I have been pleased to see Dr. Megan Milligan's research program take shape, and to start to bring back the in-house wildlife research capacity that our unit has lacked for the past few years. Her program is on an upward trajectory, punctuated by the growth in her lab over the past year. I'm glad to see her building new research partnerships while maintaining existing partnerships as well. I am heartened that AKCFWRU is again able to meet both the wildlife and fisheries sides of its mission through the combined efforts of our groups.

My own research program has grown expansively, reflected most acutely in a vibrant research group that currently stands at 10 students. It remains a privilege and an honor to mentor such brilliant early career scientists. Their interests and Alaska's research needs have allowed us to geographically expand our reach to nearly every corner of the state: from Kivalina to Metlakatla, from Katmai to Circle, and points in between. Our research has coalesced around topics of stream restoration and environmental change, including a focus on fish growth responses to mining remediation on BLM and NPS lands, shoreline habitat influences on fish presence in projects jointly managed by ADF&G and USFWS, and the impacts of "rusting rivers" on fisheries and food webs in important, mineral-rich watersheds that are also critical to fisheries subsistence use. Over the next year we will add a new research focus in fish movement and population genetics through several exciting new projects on northern pike, sheefish, and whitefish in national parks and wildlife refuges. Looking back, I have also been deeply proud to see my first cohort of four students graduate over the past year, along with a post-doc who is now starting a tenure track faculty position at an R1 university.

AKCFWRU benefits from phenomenal administrative support as well, particularly from our Unit Administrator Monica Armbruster who earned WMI's Administrative Excellence Award this year. She keeps the AKCFWRU ship running and has been instrumental in proposal development, grant management, student support, and overseeing facilities improvements that make AKCFWRU tangibly a better place to work. Through her efforts, Megan's, and my own, I continue to see accelerating growth and a bright future for AKCFWRU, and I remain grateful for the opportunity to lead this amazing unit.



Message from the Assistant Unit Leader – Wildlife

Dr. Megan Milligan

It has been quite a first year for me as Assistant Unit Leader and I'm excited to be kicking off several exciting projects, building a research team, and continuing to get to know our cooperators and state.

I am thrilled to have recruited two graduate students this year, who will be working with ADF&G and USFWS scientists on valuable research projects. Riley Marcus is studying emperor geese on their wintering grounds from Kodiak out along the Aleutian chain in collaboration with ADF&G, evaluating the effects of wintering location on annual productivity and assessing potential mechanisms behind recent population declines. He will be conducting fieldwork this winter throughout the Aleutian chain.

Arrow Myers is studying spectacled eiders in collaboration with USFWS, leveraging a long-term monitoring dataset to evaluate the effects of changing environmental conditions on survival and population dynamics. He will hopefully be conducting fieldwork next summer with USFWS on the Yukon-Kuskokwim Delta.



I have also been fortunate enough to contribute to cooperators near and far through my own research. As just some examples, I am collaborating with the Threatened, Endangered, and Diversity Program at ADF&G on research evaluating the broad-scale factors influencing population trends in Alaskan breeding birds. I am also working with the Nevada Department of Wildlife to model and map priority habitat for mule deer throughout the state.

Beyond research, I taught my first class at UAF this spring: Population Dynamics of Vertebrates. It was an exciting challenge to develop all the material for this course. I particularly appreciated the discussions with students that helped me think about my own research in different ways. I also helped develop a new course for the department: Applied Bayesian Statistics in Ecology. Finally, I am helping to organize IAB's Life Sciences Seminar, where we're excited to bring in speakers from across the state to discuss a wide range of topics.

I am looking forward to the many possibilities for research, technical assistance, and collaboration in support of AKCFWRU. One of my highest priorities is to continue to build strong relationships with our cooperators and to work together to make our state a leader in science-based wildlife management. I look forward to collaborating with many of you in the future and contributing to our shared goals regarding Alaska's wildlife and landscapes.

Unit Cost-Benefit Statements

In-Kind Support

In-kind support, usually operational support of field activities, is critical to the success of the Alaska Cooperative Fish and Wildlife Research Unit. Although the monetary value of this support is not known, a listing of this assistance is provided for each project in this report.

Benefits

- Students Graduated (advised by Unit faculty): 5
- Scientific and Technical Publications: 10
- Data and Code Releases: 2
- Presentations: 32

Courses Taught

Jeff Muehlbauer: Research Design (UAF, 3 credits, Fall 2025)

Jeff Muehlbauer: Fisheries Seminar (UAF, 1 credits, Spring 2026)

Megan Milligan: Population Dynamics of Vertebrates (UAF, 3 credits, Spring 2026)

Theses and Dissertations of Unit Graduate Students

- Hermus, J.W. 2026. The effect of thermal stress on juvenile Chinook and coho salmon growth in the Deshka River: Fairbanks, Alaska, USA, University of Alaska Fairbanks, M.S., 32 p.
- Keller, D.E. 2026. Macroinvertebrates as a biological response to restoration of placer mined streams in Interior Alaska: Fairbanks, Alaska, USA, University of Alaska Fairbanks, M.S., 84 p.
- Kwit, I.A. 2025. Exploring phenological mismatch and the role of aquatic insects for olive-sided flycatchers (*Contopus cooperi*) breeding in boreal Alaska: Fairbanks, Alaska, USA, University of Alaska Fairbanks, M.S., 35 p.
- Wilson, E.E.F. 2026. Wakes, erosion, and beach-spawning sockeye salmon (*Oncorhynchus nerka*) at an urbanizing Alaska lake: Fairbanks, Alaska, USA, University of Alaska Fairbanks, M.S., 69 p.
- Zambito, A. 2026. Investigating Sitka black-tailed deer habitat use and predator-prey interactions with camera traps on Prince of Wales Island, Southeast Alaska: Fairbanks, Alaska, USA, University of Alaska Fairbanks, M.S., 64 p.

Scientific and Technical Publications

- Cathcart, C.N., J.A. Falke, J. Fox, R. Henszey, and K. Lininger. 2024. Multiscale processes drive formation of logjam habitats and use by juvenile chinook salmon across a boreal stream network in Alaska. *River Research and Applications* 41:593–608. <https://doi.org/10.1002/rra.4387>.
- Delbecq, C., J.B. Fellman, R.J. Bellmore, E. Whitney, K. Fitzgerald, and J.A. Falke. 2026. Season and antecedent conditions impact concentration-discharge relationships for dissolved organic and inorganic carbon in a Southeast Alaskan watershed. *Journal of Geophysical Research: Biogeosciences* 131:e2025JG009090. <https://doi.org/10.1029/2025JG009090>.
- Dunkle, M.R., K.K. Bartz, P.W. C. Gabriel, S.M. Collins, and J.D. Muehlbauer. 2025. Leveraging monitoring data to model trends in lake water quality in Southwest Alaska. *Alaska Park Science* 24:36-45.
- Grabowski, T.B., Y. Tsang, D. Bartz, C. Yap, J. Falke, J.R. Bellmore, and J. Fellman. 2026. Flow ecology of invasive suckermouth catfish in urbanized ridge-to-reef systems on O’ahu, Hawai’i. *Frontiers in Environmental Science* 14:1754403. <https://doi.org/10.3389/fenvs.2026.1754403>.
- Milligan, M.C., P.S. Coates, B.G. Prochazka, M.P. Chenaille, S.T. O’Neil, S.R. Mathews, J.R. Small, K. Miller, and S. Abele. 2025. Changes in spatial distribution and abundance together determine potential for population persistence for greater sage-grouse. *Diversity and Distributions* 32:e70092. <https://doi.org/10.1111/ddi.70092>.
- Robbins, C.J., T. Simmons, and J.D. Muehlbauer. 2025. Streams as sentinels of change in central Alaska. *Alaska Park Science* 24:16-25.
- Shaftel, R., M.L. Feddern, S.A. McAfee, E.R. Schoen, C. Cunningham, V.R. von Biela, J. Paul, Y. Cheng, A. Newman, M. Perdue, J. Schwenk, A. von Finster, and J.A. Falke. 2026. Integrating

climate data and river modeling to reveal Chinook salmon habitat conditions in subarctic river basins. *Ecosphere* 17:e70399. <https://doi.org/10.1002/ecs2.70399>.

Strohm, D.D., C.J. Sergeant, J.D. Paul, and J.A. Falke. 2025. Streamflow regime characterization in the changing boreal ecosystem: wildfire impacts from stream-to-regional scales. *Science of the Total Environment* 991:179770. <https://doi.org/10.1016/j.scitotenv.2025.179770>.

Swanson, S., M. Flamme, J. Schmidt, S. Crimmins, C. Roland, and K. Kielland. 2026. Seasonal drivers of density in a subarctic population of northern red-backed voles. *Ecology and Evolution* 16:e73142. <https://doi.org/10.1002/ece3.73142>.

Toskey, E.K., T. Matsuoka, P.M. Kiffney, S.M. Naman, M.C. Liermann, J. Bailey, D.A. Beauchamp, J.R. Bellmore, C.A. Binckley, S. Blumenshine, T.J. Caldwell, M.P. Carey, S. Claeson, C. Clawson, R. Cox, K.J. Cromwell, R.J. Danehy, M.J. Dodrill, J.J. Duda, J.B. Fellman, A.Z. Grimm, T.B. Hallbert, C.E. Jordan, E.R. Keeley, B.P. Kennedy, T.A. Kennedy, A.E. Kohler, E.D. Lowery, K. Macneale, K.D. Martens, S.A. Morley, J.D. Muehlbauer, J.B. Neuswanger, A. O'Dowd, S. Pneh, M. Power, S.J. Rhoades, J. Rosenfeld, G. Rossi, B.L. Sanderson, K.K. Shelby, T. Starkey-Owens, J.N. Thompson, C. Volk, J.S. Williamshen, and R.W. Wisseman. 2026. DRIFteRS: A dataset of drift invertebrate densities in streams and rivers across western North America, 1997–2024. *Ecology* 107:e70387. <https://doi.org/10.1002/ecy.70387>.

Data and Code Releases

Coates, P.S., M.C. Milligan, S.T. O'Neil, B.E. Brussee, and M.P. Chenaille. 2025. Rasters representing Greater Sage-grouse space use, habitat selection, and survival to inform habitat management (ver. 3.0). U.S. Geological Survey data release. <https://doi.org/10.5066/P933VE6W>.

Toskey, E.K., T. Matsuoka, P.M. Kiffney, S.M. Naman, M.C. Liermann, J. Bailey, D.A. Beauchamp, J.R. Bellmore, C.A. Binckley, S. Blumenshine, T.J. Caldwell, M.P. Carey, S. Claeson, C. Clawson, R. Cox, K.J. Cromwell, R.J. Danehy, M.J. Dodrill, J.J. Duda, J.B. Fellman, A.Z. Grimm, T.B. Hallbert, C.E. Jordan, E.R. Keeley, B.P. Kennedy, T.A. Kennedy, A.E. Kohler, E.D. Lowery, K. Macneale, K.D. Martens, S.A. Morley, J.D. Muehlbauer, J.R. Neuswanger, A. O'Dowd, S. Pneh, M. Power, S.J. Rhoades, J. Rosenfeld, G. Rossi, B.L. Sanderson, K.K. Shelby, T. Starky-Owens, J.N. Thompson, C. Volk, J.S. Williamshen, and R.W. Wisseman. 2025. DRIFteRS (DRift Invertebrates For salmonids in River Systems): A dataset of drift invertebrate densities in streams and rivers across western North America, 1997–2024 ver 1. Environmental Data Initiative. <https://doi.org/10.6073/pasta/234b23f08b48458e7d3910045ebacd9f>.

Presentations

Bundschuh, M., B.M. Creel, C. Kotalik, J.M. Kraus, J.D. Muehlbauer, B.G. Perrotta, F.E. Rowland, R. Schulz, D.M. Walters, and J. Wesner. 2026. Contaminant spiraling as a unifying conceptual model for predicting fate and effects of contaminants on aquatic-terrestrial meta-ecosystems. Society for Freshwater Science Annual Meeting, Spokane, Washington, 17-21 May 2026.

Condon, L.A., M.C. Milligan, K.M. Acuña, S.T. O'Neil, and P.S. Coates. 2025. Intact habitat for greater sage-grouse includes biological soil crusts. 11th World Conference on Ecological

Restoration, Denver, Colorado, 30 September-4 October 2025. Recorded but not delivered due to shutdown.

Dunkle, M., P.W.C. Gabriel, S.R. Textor, K.K. Bartz, and J.D. Muehlbauer. 2026. Landscape drivers of lake thermal structure in Southwest Alaska's National Parks. Society for Freshwater Science Annual Meeting, Spokane, Washington, 17-21 May 2026.

Ericson, T.A. and J.D. Muehlbauer. 2026. Developing climate informed approaches for stream habitat restoration in Southeast Alaska. Alaska Chapter American Fisheries Society Annual Meeting, Fairbanks, Alaska, 23-27 March 2026.

Harings, M.A.B, E. Schoen, K. Reece, B.C. Kamermans, J. Falke, and J.A. López. 2026. Evaluating spatial and temporal patterns in environmental DNA concentrations during historically low salmon runs in a Yukon River tributary, Alaska. Alaska Chapter of the American Fisheries Society Annual Meeting, Fairbanks, Alaska, 23-27 March 2026.

Harings, M.A.B, E. Schoen, K. Reece, B.C. Kamermans, J. Falke, and J.A. López. 2026. Evaluating spatial and temporal patterns in environmental DNA concentrations during historically low salmon runs in a Yukon River tributary, Alaska. American Fisheries Society Western Division 2026 Annual Meeting, Portland, Oregon, United States, 4-7 May 2026.

Hermus, J.H., J.D. Muehlbauer, D.J. Rinella, V.R. von Biela, and J.A. Falke. 2026. Juvenile salmon thermal resilience: A look into a warming future. Alaska Chapter American Fisheries Society Annual Meeting, Fairbanks, Alaska, 23-27 March 2026.

Hermus, J.H., J.D. Muehlbauer, D.J. Rinella, V.R. von Biela, and J.A. Falke. 2026. Juvenile salmon thermal resilience: A look into a warming future. Alaska Chapter American Fisheries Society Annual Meeting, Fairbanks, Alaska, 23-27 March 2026.

House, T. and J.D. Muehlbauer. 2026. Assessing impacts of mining on macroinvertebrate prey resources. Alaska Chapter American Fisheries Society Annual Meeting, Fairbanks, Alaska, 23-27 March 2026.

House, T.F. and J.D. Muehlbauer. 2026. Assessing energetic budgets in mining impacted sub-arctic streams. Society for Freshwater Science Annual Meeting, Spokane, Washington, 17-21 May 2026.

Keller, D.E. and J.D. Muehlbauer. 2026. Cripple Creek: second and third assessment of macroinvertebrate communities. Alaska Chapter American Fisheries Society Annual Meeting, Fairbanks, Alaska, 25-27 March 2026.

Malley, B.J., P.S. Coates, M.C. Milligan, S.R. Mathews-Sanchez, and G.L. Patricelli. 2026. Between the ashes and the sage: Movement and habitat use of greater sage-grouse broods in burned and unburned landscapes. Western Section of The Wildlife Society, Monterey, California, 2-6 February 2026.

Malley, B.J., P.S. Coates, M.C. Milligan, S.R. Mathews-Sanchez, and G.L. Patricelli. 2026. Between the ashes and the sage: Movement and habitat use of greater sage-grouse broods in burned and unburned landscapes. Western Association of Fish and Wildlife Agencies 35th

Biennial Sage and Columbian Sharp-Tailed Grouse Workshop, Spearfish, South Dakota, 18-21 August 2026.

Milligan, M.C., K.S. Christie, and J.C. Hagelin. 2026. Broad-scale predictors of population trends in Alaskan breeding birds. Alaska Chapter of The Wildlife Society Annual Meeting, Anchorage, Alaska, 8-10 April 2026.

Milligan, M.C. and P.S. Coates. 2025. Drought nullifies positive effects of habitat quality on population growth for a sagebrush indicator species. The Wildlife Society's 32nd Annual Conference, Edmonton, Canada. Canceled due to shutdown.

Milligan, M.C., P.S. Coates, J.L. Beck, J.B. Dinkins, C.A. Hagen, C.P. Kirol, T. Owens, A.C. Pratt, J.S. Sedinger, K.T. Smith, J. Struthers, P.A. Street, C.L. Aldridge, and A.P. Monroe. 2026. Effects of free-roaming horses on vital rates of greater sage-grouse across their range. 16th International Grouse Symposium, Hamar, Norway, 31 August-4 September 2026.

Milligan, M.C., P.S. Coates, and S.J. Dettenmaier. 2026. From behavior to treatment: integrating ecological trap theory into conifer management decision support for greater sage-grouse. 16th International Grouse Symposium, Hamar, Norway, 31 August-4 September 2026.

Moody, E.K, J.D. Muehlbauer, P. Brown, J.R. Corman, B.R. Deemer, T.A. Kennedy, and C.B. Yackulic. 2025. Calcium carbonate precipitation influences processes across multiple trophic levels in the Little Colorado River. Desert Fishes Council. Cuatro Cienegas, Mexico, 19-23 November 2025.

Muehlbauer, J.D., S. Larsen, J. Wolfram, J. Wesner, D. Kopp, J. Kraus, J. M. Álvarez-Martínez, A. Grigoropoulou, and R. Schultz. 2025. Join the dark side: aquatic insects and global contaminant transport. Alaska Chapter American Fisheries Society Annual Meeting, Fairbanks, Alaska, 23-27 March 2026.

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***Note:** Reports are listed as Completed or Ongoing in the categories of Fisheries, Wildlife, or Ecological Studies, in that order. It is recognized that these distinctions are somewhat artificial. A list of abbreviations appears on the final page of the report. For completed studies, the abstract of the student's thesis or dissertation is provided where possible.

Completed Fisheries Studies

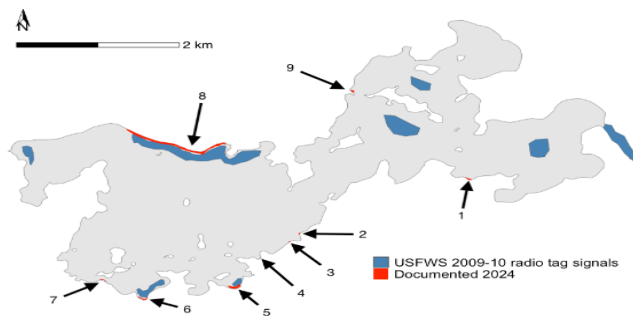
Wakes, Erosion, and Beach-Spawning Sockeye Salmon (*Oncorhynchus nerka*) at an Urbanizing Alaska Lake

Student Investigator: Eli Wilson, MS Fisheries

PI: Jeff Muehlbauer

Funding Agency: ADF&G RTS (Sport Fish Base), USGS SSP (RWO 255), UAF URSA

In-Kind Support: ADF&G, USFWS



Sockeye salmon mapped previously using tagged fish vs in this study using drone imagery.

Human development along lakeshores has the potential to impact lake habitat quality and alter natural processes occurring in those systems, including erosion and sediment transport. While much of Alaska is free of urbanization, specific areas of the state have seen extensive development in the past while others have seen an acceleration in development in more recent years. As a case study we sought to determine the frequency of recreational

watercraft wake activity and its association with shoreline erosion in Big Lake, Alaska. This lake is a key regional producer of Sockeye Salmon (*Oncorhynchus nerka*) and is located in one of Alaska's urbanizing areas, the Matanuska-Susitna Valley. I documented mean shoreline erosion rates varying by site from as little as 8 mm to as much as 92 mm over the course of the 2024 field season and found that sites with relatively high levels of wake activity saw significantly more erosion. Additionally, I mapped the spatial distribution of beach spawning activity by Sockeye Salmon within Big Lake with the aid of unmanned aerial systems (commonly known as a drone), simultaneously gathering relative abundance data for sites around the lake. I identified spawning activity at a total of nine sites, six previously unmapped, though it is unclear if these newly identified locations were missed during previous survey efforts or reflect interannual variation in spawner abundance or habitat changes since the last surveys. Finally, I assessed between-counter variation in relative abundance estimates when using drone footage. Quantification of variation between counters has long been an issue for alternative methods that require counts to be collected live in the field. I found that inter-observer variation was low (median CV = 19%) for drone footage analysis. Across this thesis, the implementation of new and developing technologies ultimately allowed me to collect data on both wake activity, for which the cost of traditional methods would have been prohibitive, as well as the relative abundance of Sockeye Salmon via a potentially lower-cost, more effective method.

The Effect of Thermal Stress on Juvenile Chinook and Coho Salmon Growth in the Deshka River

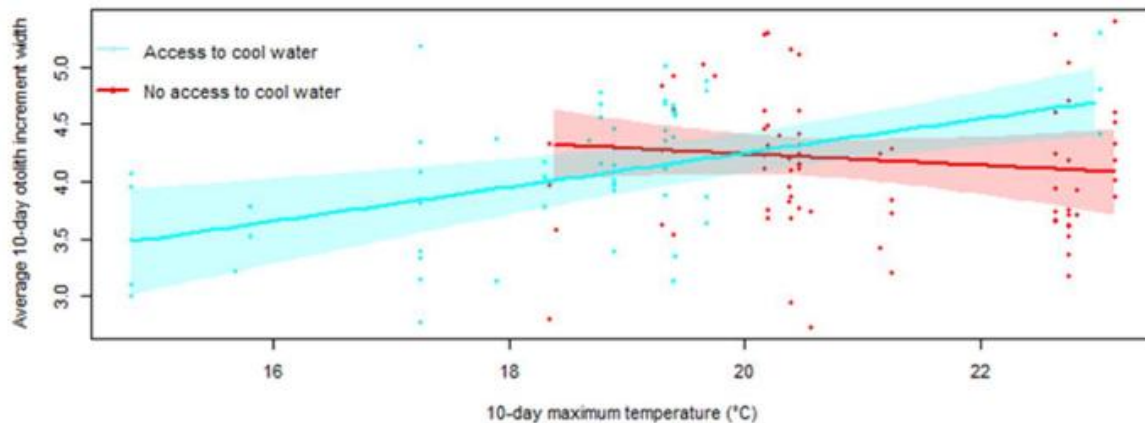
Student investigator: John Hermus, MS Fisheries

PI: Jeff Muehlbauer

Funding: USGS SSP (RWO 256)

In-Kind Support: USFWS

Warming climate and declines of salmon, cold-water fishes, across the State of Alaska have raised concerns about the future of Alaskan salmon populations. This study examines sublethal effects of warm freshwater temperatures during juvenile rearing of two co-occurring Pacific salmon species by linking water temperatures to growth rates. Juvenile growth of Pacific salmon plays a key role in their success because individuals that enter the ocean at larger sizes tend to have higher survival rates. We sampled juvenile Chinook salmon and Coho salmon in the Deshka River, where summer water temperatures routinely exceed 20 °C, likely making it one of the warmest known salmon streams in Alaska. During the summers of 2019–2022, juvenile salmon were trapped at pre-established temperature monitoring sites throughout the watershed. A subset of these fish were lethally sampled to obtain otoliths for daily growth analysis, stomachs for fullness, and muscle tissue for heat stress determination. The results of this study suggest juvenile Chinook and Coho salmon are capable of high growth rates up to approximately 23 °C, implying that food availability has kept up with the increases in metabolic demand associated with these high-water temperatures. Beyond this threshold, water temperatures approach their upper incipient lethal temperature, a thermal limit at which a majority of fish will die after prolonged exposure of seven days or more. Our findings demonstrate that thermal heterogeneity is vital for supporting juvenile salmon growth, with access to cool-water refugia enhancing growth rates under warm water conditions of 19–23 °C. This underscores the importance of conserving river connectivity and natural flow regimes that sustain access to essential thermal refugia.



Juvenile Chinook salmon growth for fish with and without access to cooler water refugia.

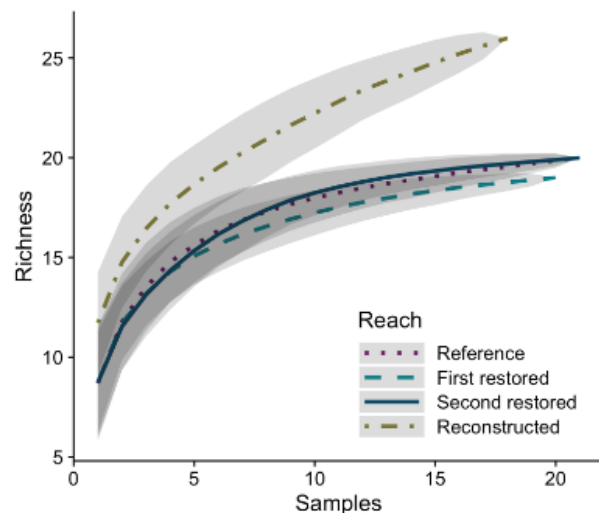
Macroinvertebrates as a Biological Response to Restoration of Placer Mined Streams in Interior Alaska

Student Investigator: Dakota Keller, MS Biological Sciences

PI: Jeff Muehlbauer

Funding Agency: BLM (RWO 258), IAB, UAF URSA, Golden Valley Electric Association, Ascentra Credit Union, DBW Nicholas Hughes Scholarship, Calvin J. Lensink Endowment

In-Kind Support: DBW, USFWS, BLM, TVWA, IALT, AFS Hutton Scholars



A caddisfly under the microscope from one of the Cripple Creek macroinvertebrate samples. Photo credit: Dakota Keller.

To return ecosystem function to formerly placer mined streams in Interior Alaska, restoration approaches are utilized to expedite floodplain revegetation, stabilize stream channels, and increase pool formation through boulder and large woody debris additions. Despite decades-long use of these techniques to restore habitat, a disconnect still exists between the restoration approaches we use and the biological outcomes we often desire, such as measurable improvements to fisheries. To better understand the biological efficacy of stream restoration techniques, I measured benthic macroinvertebrates in two formerly placer mined streams, Cripple Creek and Nome Creek, that have undergone restoration to improve fish habitat. Both streams were

sampled for macroinvertebrate communities within their initial recovery phases (1–3 years post restoration) by comparing restored reaches with reaches disturbed by mining and urban development. Because Cripple Creek’s restoration utilizes a historic channel with existing natural habitat, I expected higher abundance and biomass within the restored reach compared to the unrestored reaches. Similar richness and diversity were anticipated in all Cripple Creek reaches due to recolonization by drifting macroinvertebrates. In Nome Creek, I expected the lowest indices of abundance, richness, diversity, and biomass to be found within reaches most recently disturbed by restoration construction or historic mining. For both creeks, I instead found model support for similar macroinvertebrate metrics across restored and unrestored reaches, indicating rapid recovery from the disturbance of reconstruction, but not necessarily to a macroinvertebrate assemblage beyond the existing community within the unrestored reaches previously disturbed by mining or development. However, in Nome Creek, I unexpectedly found the highest biodiversity within the reach most visually altered by mining. While this reconstructed reach is not a template to be applied throughout Nome Creek, its higher biodiversity is evidence that novel, constructed channels have potential to increase macroinvertebrate productivity. Thus, further research is warranted to identify the physical characteristics of this stream reach that drive these desired biological responses to improve restoration techniques.

Evaluating Environmental DNA as a Complementary Tool for Estimating Salmon Abundance in the Yukon River Basin

Student Investigator: Maggie Harings, MS Fisheries

PI: Andrés López and Erik Schoen

Funding Agency: USGS AK CASC, NSF GRFP, NIWR WERC

In-Kind Support: ADF&G, TCC, USFWS, BLaST, UAF URSA



Maggie Harings filters river water in the Chena River to collect salmon DNA in the system. Photo credit: Erik Schoen.

Monitoring the abundance of Chinook (*Oncorhynchus tshawytscha*) and Chum (*O. keta*) salmon in the Yukon River Basin is critical for fisheries management, but declining populations and frequent high-flow events increasingly challenge conventional counting methods. High water can render salmon counting infrastructure inoperable, resulting in missed daily salmon counts and, in some cases, missed estimates of spawning abundance for a full season. To address these challenges, we evaluated the use of environmental DNA (eDNA) and environmental parameters to estimate salmon abundance in three Yukon River

tributaries. We developed a field-friendly protocol suitable for field sites lacking freezers or dedicated staff with prior molecular biology expertise and applied it at three tributaries during the historically low salmon runs of 2021–2022. No contamination was observed in negative controls collected during field sampling, including at remote weirs where personnel regularly handled salmon and may have residual DNA on clothing and waders. Trends in DNA concentrations in samples following undulations in fish passage generally lagged peaks in fish counts by 1–5 days. Generalized linear models predicting daily salmon passage from daily eDNA concentrations and gage height had generally low predictive power ($R^2_{McF} = 0.02\text{--}0.17$), likely due to variable time lags between fish passage and peak eDNA detection. The site with the highest Chum Salmon counts and eDNA concentrations exhibited the best model fit, suggesting this technique may be most suitable for estimating daily passage of Chum Salmon during years with higher salmon abundance. Additional years of paired eDNA data and salmon counts are needed to test the utility of an eDNA-based approach for estimating total annual spawning abundance, a more important metric for management applications. Together, these findings suggest that eDNA sampling is feasible at remote fisheries field camps and is unlikely to produce reliable daily salmon counts when sampling during historically low salmon runs but may complement traditional approaches for estimating annual salmon escapement in larger rivers with relatively low salmon densities, potentially supporting more robust fisheries management under changing environmental conditions.

Ongoing Fisheries Studies

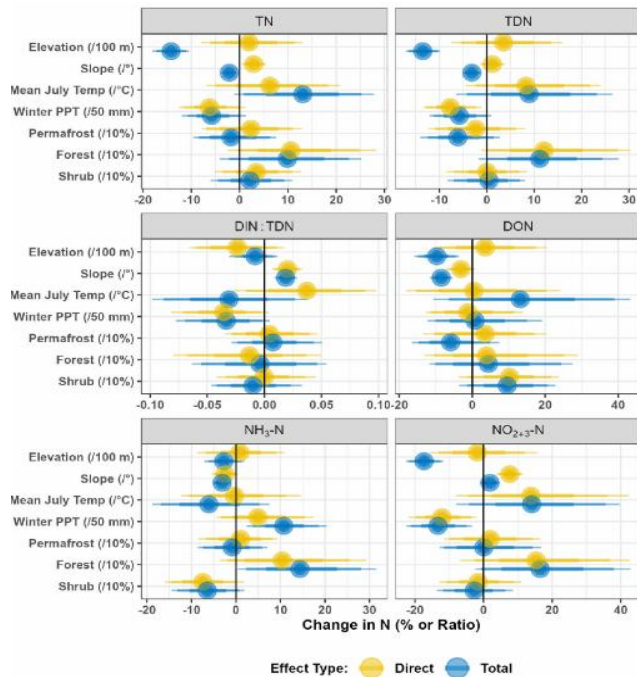
Spatial and Decadal-Scale Temporal Changes in Water Chemistry and Macroinvertebrates in Central Alaska

Collaborator: Caleb Robbins, Baylor University

PI: Jeff Muehlbauer

Funding Agency: NPS (RWO 254), BLM (RWO 258)

In-Kind Support: NPS



Modeled effects of environmental covariates on concentrations of various nitrogen forms.

Northern latitudes are disproportionately warming due to climate change, with potential consequences cascading to aquatic life. To understand and manage climate-driven changes to its resources, staff led by Trey Simmons at NPS have been collecting water chemistry (carbon, nitrogen, phosphorus, cations, pH, etc.) and benthic macroinvertebrates from ~225 streams in the Central Alaska Inventory and Monitoring Network (parks and preserves: Denali, Wrangell-St. Elias, and Yukon-Charley Rivers) since 2004. Investigator Robbins explored catchment-scale spatial drivers and decadal-scale temporal trends in the water chemistry and invertebrate communities. Modeling involved generalized linear mixed models fit in a Bayesian framework to account for site-level replication, non-Gaussian response families, censoring (water chemistry), and

measurement error (invertebrate richness). Carbon and nitrogen analyte concentrations typically decreased with increasing elevation, and phosphorus and nitrogen have tended to increase through time. Woody vegetation and permafrost also had distinct controls on various forms of N, and nitrate concentrations generally increased from late spring to early fall. The most striking and consistent (across temporally replicated sites) changes were specifically in total dissolved nitrogen, often from near detection limits (<0.005 mg N/L) to 0.2 - 0.4 mg N/L within two decades. Elevation was also a dominant catchment-scale spatial correlate of invertebrate diversity. Richness increases at individual sites ranged from 0 to a doubling of taxa (e.g., from ~20-40) within two decades. There are likely multiple mechanisms driving benthic community change across the Central Alaska Network; in some sites, there are no changes in diversity, but there are significant, consistent changes in community structure (dominance). These analyses establish baselines for monitoring in a rapidly changing Alaskan climate, as well as inform research priorities for understanding and forecasting future change. This work has resulted in a publication in a special issue of *Alaska Park Science* as well as forthcoming publications in a NPS book and peer-reviewed journal.

The Influence of Stream Flow Patterns on Juvenile Salmon Foraging and Growth in Southeast Alaska

Student Investigator: Blake Toney, MS Fisheries

PI: Curry Cunningham

Funding Agency: USGS AK CASC (RWO 244)

In-Kind Support: USFS PNWRS, UAS, ACRC



Diet sample from a Dolly Varden char containing pink salmon eggs in Montana Creek near Juneau, AK. Photo credit: Brennen McCullogh.

Climate driven shifts in Southeast Alaska's hydrology are altering the freshwater conditions critical to wild salmon populations. Regional climate models project more frequent and intense summer precipitation events, producing extreme flow conditions in salmon-bearing streams. These changes may disproportionately affect Pacific salmonid species such as juvenile coho salmon and Dolly Varden char that rear in freshwater for extended periods before migrating to the ocean. Our study investigates the role of small, second-order tributaries to provide important habitat and foraging opportunities during high-flow events in coastal river systems. Previous research suggested that tributaries can serve as high-flow refugia by offering juvenile salmonids shelter from the energetic cost of mainstem conditions during peak discharge. However, increased turbidity

and accelerated water velocity during high-flow events complicates active fish sampling, challenging past efforts to quantify the ecological value of intermittently accessible tributary habitats in coastal Southeast Alaskan streams. We used Passive Integrated Transponder (PIT) tags and portable antenna arrays to monitor juvenile coho salmon and Dolly Varden char movement between two intermittent second-order tributaries and adjacent mainstem habitats to identify whether habitat use patterns influence growth potential. Preliminary results indicate that fish moving between habitats experience significantly higher growth across the summer season than individuals exhibiting strong site fidelity. To identify potential mechanisms, we paired movement data with high-frequency diet sampling designed to capture variability in food subsidies across flow events. Fish in tributaries showed significantly higher beta diversity in diet composition and consumed a greater proportion of terrestrial prey. The proportion of terrestrial food subsidies increased with streamflow, suggesting that hydrologic variability enhances access to alternative food resources. Together, these findings provide empirical evidence that access to, and intermittent use of, tributaries expands growth potential in native salmonids by offering heterogeneous habitat and distinct energetic pathways. This research highlights the importance of tributary habitat in supporting population-level resilience under changing hydrologic regimes and underscores the need for management strategies that account for the ecological value of small, intermittently accessible tributaries.

Climate and Landscape Influences on Lake Vertical Structure in Southwest Alaskan National Parks

Post-Doc: Matt Dunkle

PI: Jeff Muehlbauer

Funding Agency: NPS (RWO 259, 266)

In-Kind Support: NPS



Sampling water clarity with a Secchi disk on Hammersley Lake, Katmai National Park. Photo credit: Matt Dunkle.

Climate change is reshaping the physical and chemical structure of subarctic freshwater ecosystems, with significant consequences for aquatic species such as Pacific Salmon. In southwest Alaska, lakes serve as integrators of watershed processes and sentinels of environmental change in temperature, chemistry, and light availability. This project leverages a long-term, spatially extensive dataset of vertical water quality profiles collected through the National Park Service Inventory & Monitoring program, in collaboration with Krista Bartz, Sadie Textor, Paul Gabriel (Alaska Dept.

Environmental Conservation), Peyton Thomas (University of Colorado Boulder), and Sarah Laske (USGS Alaska).

Since 2008, four accessible lakes have been sampled annually, while dozens of smaller, more remote lakes across Lake Clark and Katmai National Parks have been sampled approximately once every three years. Vertical profiles—now spanning nearly two decades across 44 lakes—have been collected using a YSI ExoSonde 2 multi-sensor sonde, measuring temperature, dissolved oxygen, pH, conductivity, turbidity, and ancillary conditions such as wind speed and Secchi depth.

The goal of this work is to assess long-term changes in lake thermal structure, with an emphasis on habitat suitability for rearing Sockeye Salmon and Lake Trout. We grouped lakes into three types—Large/Lowland, Small/Shallow, and Icy/High-Elevation—by morphometry and watershed characteristics, then fit LASSO-selected generalized additive mixed models to identify drivers of epilimnion and hypolimnion temperature. Summer air temperature was the strongest predictor of surface warming (epilimnion $R^2 = 0.70$), but watershed glacier cover moderated that sensitivity ($r = -0.72$ vs. glacier cover); hypolimnion temperature stayed largely decoupled from surface forcing. By 2100, projected epilimnion warming ranges from $+1.7$ – 3.6°C under moderate emissions to $+9.3$ – 10.7°C under high emissions, with Small/Shallow and Large/Lowland lakes projected to routinely exceed the $\sim 18^\circ\text{C}$ aerobic-scope threshold for rearing Sockeye by mid-century under the latter. Because the most buffered lakes are losing glacier cover fastest, cold-water refugia for salmon and lake trout may decrease. This work is in preparation as a manuscript to be submitted for peer review before January 2027.

The Natural History of Lampreys in Alaska: Distributions, Adult Migration Ecology, and Genomic Variation

Student Investigator: Nate Cathcart, PhD Fisheries

PI: Trent Sutton, CFOS and Jeff Muehlbauer

Funding Agency: BIA

In-Kind Support: ADF&G, Knik Tribe, CRITFC



Juvenile Pacific Lamprey. Photo credit: Nate Cathcart.

Lampreys in Alaska include the Arctic Lamprey (*Lethenteron camtschaticum*), Pacific Lamprey (*Entosphenus tridentatus*), and Western River Lamprey (*Occidentis ayresii*). They represent the state's most understudied yet widespread anadromous fishes. Although their ranges overlap with Pacific salmon, the distributions and ecologies of these species remain poorly understood. Without better knowledge of lamprey natural history, Alaska's resource managers will not be able to understand how or where lamprey species and life stages respond to environmental gradients

nor mitigate potential negative effects of development on lamprey populations. The primary goals of this project are to improve maps of lamprey distributions, quantify adult riverine migrations relative to ecological correlates, and measure multi-scale genomic variation of Pacific Lamprey within watersheds, throughout Alaska, and across their entire range. These goals will be met by integrating local and historical knowledge with larval, juvenile, and adult lamprey collections or observations from the field (e.g., electrofishing, fish wheels, sonar footage). Further, lamprey tissue from field collections will be genetically analyzed to measure kinship, explore life history expression, and compare genetic diversity. Although the work is statewide, the focus is on southcentral Alaska in watersheds of Cook Inlet and the Copper River basin. This project will provide a roadmap for future conservation, research, and potential management of Alaska's lampreys.

Rusting Rivers in Northwest Alaska Impacts on Stream Biota

Student Investigator: Lauren Yancy, MS Fisheries

PI: Jeff Muehlbauer

Funding Agency: USGS AK CASC (RWO 262)

In-Kind Support: ADF&G



Rusting river in the Wulik River drainage. Photo credit: Lauren Yancy.

The Arctic is experiencing rapid warming, which is altering seasonal weather and, in turn, influencing hydrologic processes. In the Alaskan Arctic, continuous permafrost is thawing and expanding the Earth's active layer creating new hydrologic connections that are mobilizing previously sequestered heavy metals. The resulting heavy metal seeps are introduced into previously pristine Arctic streams, creating a visually striking phenomenon we call "rusting rivers" due to the high iron content. Over the past decade, rusting rivers have been increasing in frequency, raising concern for the health

of the stream ecosystem, many which support subsistence and sport fisheries. This project leverages an existing biomonitoring program conducted by ADF&G in the Wulik River drainage, and introduces a new component focused on documenting the cellular health of juvenile Dolly Varden in rusting rivers. The project goals are to: (1) characterize spatial and temporal patterns in biological metrics associated with existing and newly observed rusting seeps, and (2) document histopathological impacts of rusting rivers on juvenile Dolly Varden. Results show that any level of rusting influence was associated with declines in primary production and benthic macroinvertebrate communities, while no significant decline was observed in drifting macroinvertebrate abundance and richness or juvenile Dolly Varden catch per unit effort. Preliminary histopathological findings show multifocal epithelial hyperplasia in the gills, with severe lesions resulting in lamellar fusion and loss of respiratory surface area. Notably, some fish with no observed external abnormalities still exhibited hyperplasia in the gills.

Habitat Utilization of Juvenile Chinook Salmon and Impacts of Rehabilitation Structure in the Kenai River

Student Investigator: Danielle Tryon, MS Fisheries

PI: Jeff Muehlbauer

Funding Agency: ADF&G (Sport Fish Base), PSMFC, UAF URSA

In-Kind Support: ADF&G



Minnow traps pre-deployment on the Kenai River. Photo credit: Danielle Tryon

Chinook salmon populations are under severe decline throughout their range. Many of the remaining wild Chinook salmon stocks are located in Alaska, yet even here many populations are experiencing low returns or extirpation. Many factors, from environmental stressors to fishing pressures, affect these fisheries, making it challenging to attribute declines to a single cause. One river system experiencing substantial declines is the Kenai River, located in Southcentral Alaska,

USA. The Kenai River system was once a Chinook salmon fishing destination; however, since the late 1990s there have been steady declines in abundance and size of fish, leading to fisheries closures since 2020. While substantial research effort has been focused on adult returns, little research has been conducted on the juvenile, freshwater life stage, which is a critical time that drives the subsequent success of individuals and populations later in life. Our research investigates freshwater rearing habitat availability and suitability during the summer. This study aims to document what habitats juveniles are using and to explore how riverbank rehabilitation throughout the Kenai impacts juvenile Chinook salmon habitat availability. We conducted season-long minnow trapping and habitat classification throughout the lower Kenai River from Skilak Lake to Cook Inlet. Two years of sampling will be used to model habitat suitability and preference and then be validated by additional minnow trapping efforts from ongoing projects. Early results from the first of two summer seasons demonstrate that vegetation cover, river depth, substrate size, discharge, and shoreline development are the largest predictors of juvenile habitat preference. Further research and additional data will investigate the implications of habitat preference in this developed river to better understand the influence of rehabilitation and human impacts on rearing habitat. Additionally, we hope the findings from this work will inform management recommendations regarding preferred rehabilitation structures for supporting juvenile Chinook salmon rearing habitat in future rehabilitation efforts.

Increasing Habitat Restoration Efficiency and Effectiveness in Alaska Through the Development of Geospatial Tools

Student Investigator: Tim Ericson, MS Fisheries

PI: Jeff Muehlbauer

Funding Agency: USFWS

In-Kind Support: USFWS



Klawock Indigenous Stewards Forest Partnership collects a road/stream crossing survey along Klawock Lake. Photo credit: Tim Ericson.

Southeast Alaska supports a culturally and commercially significant salmon fishery adjacent to the Tongass National Forest, home to thousands of miles of anadromous streams across the more than 1,000 islands that make up the Alexander Archipelago. However, many streams and rivers in this region have become impassable to salmon since the industrial logging boom in the 1950s fragmented fish-bearing streams with thousands of road/stream crossings as logging roads cut through low gradient coastal forests. This process continued after the Alaska Native Claims Settlement Act in 1971, as Native Corporations

continued logging on their appointed private lands, clear cutting trees while increasing private timber roads across Southeast Alaska. Inventories of road/stream crossings are incomplete, non-existent, or compiled with out of date data to currently assess fish passage along many private logging roads.

Since 2021, USFWS, local work crews, and private land owners have surveyed more than 1000 road/stream crossings across Southeast private lands to address this data gap. Utilizing this new private lands dataset, this thesis seeks to (1) develop a roadmap document to guide fish passage assessment and restoration opportunity identification across unsurveyed lands with the use of remotely sensed data, and (2) identify the highest priority fish passage projects by creating dendritic connectivity indices for the region, pairing private and public fish passage databases with remotely sensed synthetic watersheds. Collectively, this thesis will develop and validate a remote-sensing-based approach for assessing fish passage on unsurveyed private lands in Southeast Alaska, then apply that approach at a regional scale to quantify how road/stream crossings fragment habitat for anadromous and resident fishes across land ownership types.

Linking Macroinvertebrate Communities and Fish Bioenergetics in a Mined Subarctic Stream

Student Investigator: Thomas House, MS Fisheries

PI: Jeff Muehlbauer

Funding Agency: NPS (RWO 263)

In-Kind Support: NPS, CFOS



Drift nets deployed in Yukon-Charley Rivers National Preserve.

Photo credit: Thomas House.

Coal Creek, a former placer-mined stream on the upper Yukon River approximately 60 miles upstream from Circle, Alaska, provides an opportunity to examine the long-term ecological consequences of placer mining more than 50 years after mining activity ceased. Beyond its cultural and historical significance, the watershed represents potential rearing habitat for juvenile Chinook salmon (*Oncorhynchus tshawytscha*), whose populations have declined substantially throughout the Yukon River drainage. Historical mining dramatically altered the physical structure of Coal Creek, with potential consequences for the aquatic macroinvertebrate communities that form an important component of the prey base supporting juvenile fish growth.

Previous sampling in the watershed has indicated greater macroinvertebrate abundance and diversity in nearby unmined reference streams relative to Coal Creek, suggesting that effects of mining may persist for decades following disturbance. My research is evaluating how these legacy impacts influence the availability and energetic quality of prey for rearing fishes. To accomplish this, my team and I collected macroinvertebrate and fish diet data from Coal Creek and two nearby unmined reference streams, Sam Creek and Thanksgiving Creek. Macroinvertebrate availability was characterized using Surber and drift sampling, while gastric lavage samples were collected from fishes to evaluate prey selection. Water quality, stream temperature, and riparian habitat characteristics were also measured to provide environmental context for observed differences among reaches. Ultimately, measurements of macroinvertebrate abundance and energy density will be used to parameterize a fish bioenergetics model and explore differences in potential juvenile salmonid growth among mined and reference streams.

Field sampling for the study is now complete, and laboratory processing of the collected macroinvertebrate samples is well underway. Approximately one-third of the samples have been processed to date, with the remaining laboratory work expected to be completed by the end of September 2026. Using preliminary data I was able to give talks at the Alaska Chapter of the American Fisheries Society's annual conference in Fairbanks as well as at the Society for Freshwater Science's annual conference in Spokane. Data analysis and interpretation will continue throughout the fall, including comparisons of macroinvertebrate communities, fish diets, and modeled growth potential across study reaches. These analyses will form the basis of my master's thesis and will help clarify the extent to which legacy placer mining continues to influence stream food webs and the capacity of altered habitats to support rearing fishes in the Yukon River basin. I expect to defend my thesis early in the Spring 2027 semester and graduate at the end of the term.

Seaward Migration and Overwintering Habits of Dolly Varden in Northwestern Alaska Analyzed Using Otolith Microchemistry

Student Investigator: Joseph Spencer, MS Fisheries

PI: Andy Seitz, CFOS

Funding Agency: ADF&G Region 3 (Sport Fish Base), USFWS OSM

In-Kind Support: ADF&G, USFWS



Packrafting the upper Noatak River, Alaska.

Diadromous Dolly Varden are an important subsistence resource in northwestern Alaska. They feed in the ocean during ice-free months and rear, spawn, and overwinter in rivers. Frequent inter-river movements for overwintering and spawning are suspected, but life history and migratory strategies of populations across the region are not well understood. Movements among rearing, spawning, and overwintering habitats expose Dolly Varden to numerous subsistence, commercial, and sport fisheries.

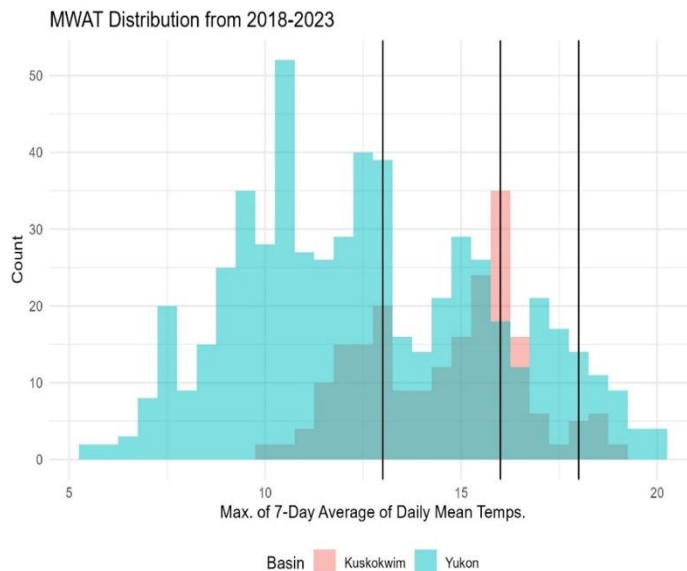
Knowledge of migratory strategies is needed to fully understand population dynamics, exposure to harvest, and impacts of climate change. The objective of this study was to elucidate the migration strategies of Dolly Varden populations using otolith microchemistry. Between 2021 and 2024, we collected and analyzed otoliths from Dolly Varden captured ($n=295$) in spawning areas of northwestern Alaska, from subsistence fisheries ($n=91$) in Kivalina and Noatak villages, and from bycatch in the commercial salmon fishery ($n=31$) in Kotzebue. Strontium concentrations measured by laser ablation along the growth axis of otoliths were used to determine the age at first seaward migration, frequency of seaward migrations, and movement between overwintering sites. Results indicate that Dolly Varden in the region typically smolt at age 3 and undergo 3–4 ocean migrations prior to spawning. Among spawners, 90.5% skipped at least one ocean migration during their lifetime, and repeat spawners comprise an average of 29% of the spawning population. This research informs management of fisheries in the region and establishes information about life history and migratory strategies to track changes in a rapidly warming Arctic.

Downscaling Recent and Future Freshwater Processes Across Two Large Subarctic Watersheds: Assessing Effects on Freshwater Habitat Potential and Productivity

Student Investigator: Rebecca Shaftel, PhD Fisheries

PI: Jeff Falke, IAB

Funding Agency: USGS AK CASC (RWO 246), USFWS R&M (RWO 257)



Model-predicted maximum weekly stream temperatures for sites across the Yukon and Kuskokwim watersheds for a recent 5-year period. Vertical lines at 13°C, 16 °C, and 18 °C indicate upper optimal temperatures for spawning, rearing, and

Alaska is experiencing climatic change faster than any other area of the United States. There is growing concern that warming stream temperatures and extreme events may be negatively impacting adult salmon, while effects on other life stages or other fish species are unknown due to limited monitoring data and a lack of stream temperature information. The objective of this project is to model stream temperatures under current and future climate scenarios across the Yukon and Kuskokwim River basins at the stream reach (~ 1 km) scale. Modeled stream temperatures from the Yukon basin will be used in a life cycle model to explore impacts on Chinook and chum salmon life stages.

Stream temperatures were modeled using boosted regression trees. Inputs to the model included gridded air temperatures and modeled streamflow in addition to spatially explicit hydrologic and topographic variables tied to the stream reach and watershed. The model was validated using data from 18 sites and two years withheld from the training data. Testing RMSE was 1.21°C and MAE was 0.91°C. Predictions will be applied across stream reaches in the study area and used to evaluate historic thermal suitability for salmon migration, spawning, and rearing habitats. Newly generated downscaled climate model projections from CMIP6 will be used to evaluate future stream temperatures across the Yukon and Kuskokwim while accounting for model and scenario uncertainty.

A recently developed life cycle modeling framework (Bellmore et al. 2022) will be used to examine the effects of short-duration stream temperature and streamflow events, along with other well-established drivers such as density-dependence, on salmon population survival through time. The goal of this project will be to develop model parameters (e.g. spawning timing, marine mortality, etc.) that are specific to summer and fall chum and Chinook salmon populations in the Yukon River Basin and examine scenarios of warming stream temperatures and high streamflow events on population abundances over a multidecadal simulation.

Understanding Life History and Growth Dynamics of Black Rockfish in Southeast Alaska During a Period of Rapid Fishery and Environmental Change

Student Investigator: Isaac Nyameke, PhD Fisheries

PI: Peter Westley, CFOS

Funding Agency: ADF&G Region 1 (Sport Fish Base), NSF Tamamta, UAF Troth Yeddha'

Black rockfish (*Sebastes melanops*) is one of 33 long-lived rockfish species found along the coastlines of Alaska. It is considered an economically important sport harvest species. However, restrictions on salmon and halibut fisheries have led to increased black rockfish harvests in recent years. This has raised concerns among coastal communities and scientists about the potential for overfishing and long-term sustainability. To address this knowledge gap, my study aim to better understand changes in life-history patterns and growth in black rockfish in Alaska's Southeast region, using a continuous demographic dataset from sport-caught black rockfish in Sitka from 2016 to 2024 with three key objectives: (1) track cohort trends over time, quantify changes in mean length, age, and body condition (Le Cren'-(Kn), (2) estimate growth (increase in length with age as a proxy for growth) and compare variation within and among years and between sexes and (3) explore the effect of air temperature (as a proxy for sea surface temperature) on young and adult growth.

Those objectives guided the research questions and hypotheses. To address Objective 1, I used descriptive statistics and generalized additive models to evaluate interannual mean trends over time while accounting for sex-specific and seasonal effects. To estimate growth and evaluate the effect of temperature (Objectives 2 and 3), I modeled fork length as the response variable using both parametric and non-parametric approaches such as Von Bertalanffy Growth Functions (VBGF) to estimate growth parameters (L_{∞} , K , and t_0) by year groupings and temperature regimes (warm vs. cool years).

Our preliminary results revealed an age truncation in the recent harvest relative to the last 10 years, driven by current intensive fishing pressure (Figure 1). Females were observed to grow faster and larger in size compared to males (Figure 2), while the younger fish had a faster growth rate in the warm years than the adults. We anticipate younger fish to convert more energy into growth, while adults use that energy for reproduction and larval development. The findings have improved our understanding of black rockfish life-history traits and growth dynamics in Southeast Alaska.

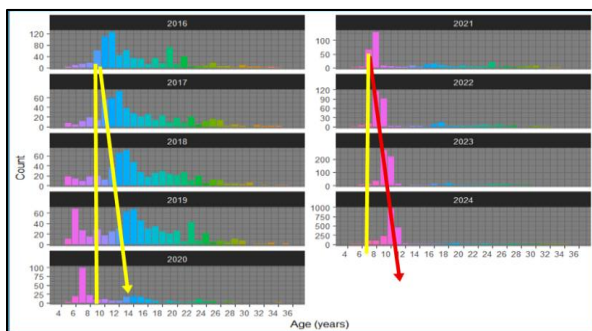


Figure 1. Age truncation in Sitka black rockfish harvest.

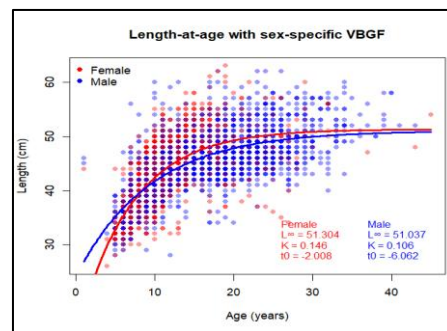


Figure 2. Female and male growth difference.

Completed Wildlife Studies

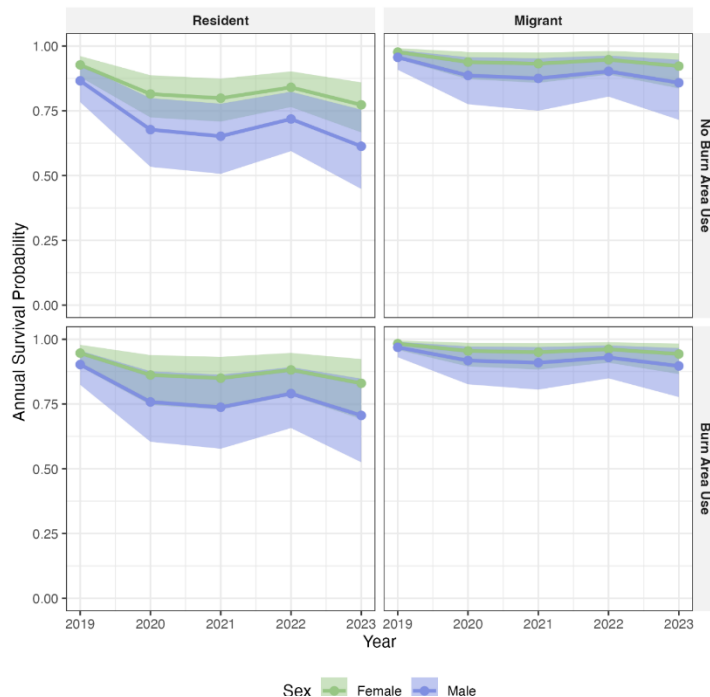
Big Game Movement and Vital Rate Modeling

Post-Docs: Alison Ketz, PhD, Derek Arnold, PhD

PI: Shawn Crimmins, IAB

Funding Agency: ADF&G (RSA G00015954)

In-Kind Support: Data provided by ADF&G



Survival probability of adult moose in southcentral Alaska, 2021-2023, as a function of sex, migratory status, and use of recently burned habitat.

Preliminary results suggest that moose which displayed migratory behavior and used a prescribed burn area had lower hazard (i.e. higher survival). Moreover, female moose had a lower baseline hazard than did male moose, and higher hazards were associated with higher levels of precipitation, potentially indicating a negative effect of rain-on-snow events. A manuscript related to this survival modeling is currently in review.

We have developed multi-scale models of moose space-use as a function of both habitat and predator occurrence, with our results suggesting that female moose with calves alter their space-use patterns during the neonatal period to avoid risk of predation, but that this alteration is scale-dependent and attenuates as calves mature. A manuscript related to this analysis is currently in review.

In recent years ADF&G completed a large-scale moose radio-collaring project in the Alphabet Hills area of southcentral Alaska. This project represents a substantial source of spatial and demographic information for moose in this region, which is of particular interest due to previous habitat management actions for improving moose forage. In collaboration with ADF&G, we will develop habitat, movement, and survival models for moose from this existing dataset in order to better understand the demography and habitat associations of moose in this region.

We used a Bayesian time-to-event framework to assess how environmental, demographic, and behavioral factors affected survival.

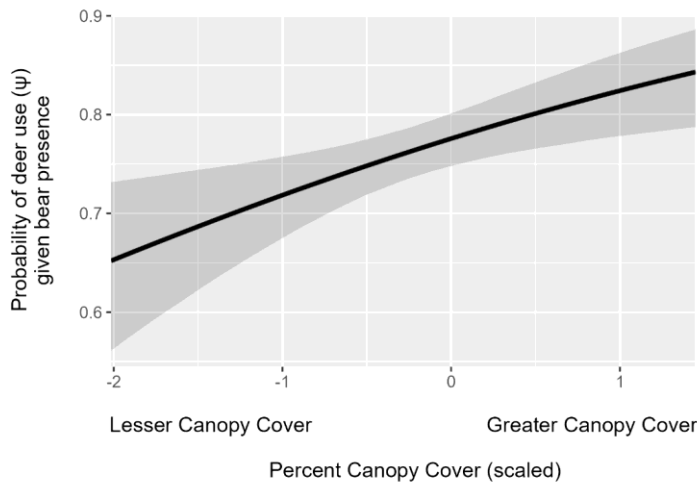
Black-Tailed Deer Density, Habitat Use, and Predator-Prey Dynamics

Post-Docs: Tony Zambito, MS Wildlife Biology and Conservation

PI: Shawn Crimmins, IAB

Funding Agency: ADF&G

In-Kind Support: ADF&G, DBW



Deer occupancy probability as a function of forest canopy cover when black bears are present.

Effectively monitoring Sitka black-tailed deer in southeast Alaska is difficult due to inaccessibility and heavy forest cover, making effective management actions challenging. Recent listing petitions for wolves in this region also necessitate an effective approach to monitoring deer, which are an important prey species for wolves. Deer are an important game species in southeast Alaska, and a critical driver of predator population dynamics in the region. On Prince of Wales Island (POW),

where habitat management for deer is a major land use issue, it is critical to understand where deer occur on the landscape and how they respond to timber harvest. Our understanding of how predators shape deer distribution on the landscape could also have important implications for harvest and habitat management. The objective of this study is to build on previous research that has demonstrated that deer respond to habitat management in terms of browsing pressure by also linking overall occupancy rates and interactions with predators to habitat conditions. We used motion-capture and time-lapse images from cameras deployed in summer of 2023 to develop single-season and multi-species occupancy models for deer. These models indicated that deer habitat use is affected by the presence of predators but that this is mediated by habitat conditions and varies seasonally. The potential for non-invasive multi-species monitoring frameworks could provide a cost-effective mechanism for assisting with harvest and/or habitat management decisions. Additionally, this project provided information on the ecological relationships between predator species and deer in southeast Alaska. A manuscript resulting from this research is in development and will be submitted in the next year.

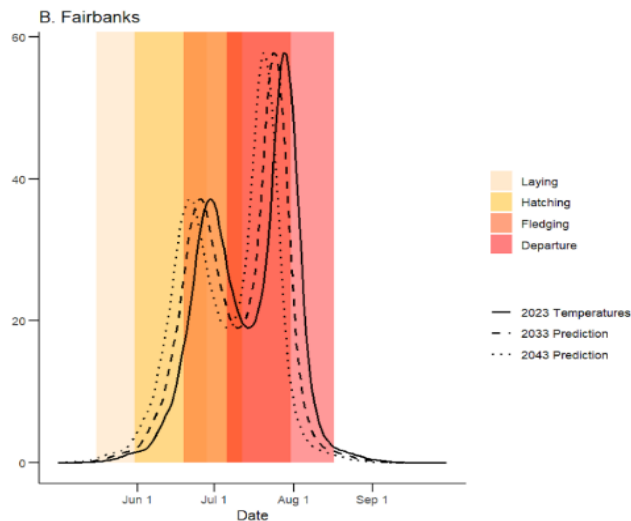
Exploring Phenological Mismatch and the Role of Aquatic Insects for Olive-Sided Flycatchers (*Contopus cooperi*) Breeding in Boreal Alaska

Student Investigator: Ian Kwit, MS Biological Sciences

PI: Jeff Muehlbauer

Funding Agency: ADF&G (Wildlife Base), NSF EPSCoR, IAB

In-Kind Support: INBRE, ADF&G, VetMed, DBW



Modeled daily emergence compared with regional *C. cooperi* breeding schedules in Fairbanks.

Migratory insectivorous birds are an increasingly threatened group worldwide. These birds rely on an ephemeral resource of insects during a short window in which they arrive at breeding territories to rear young. Climate change has the potential to create a phenological mismatch between the timing of bird breeding and the availability of key insect prey. This study investigates the potential for such a phenological mismatch in a population of olive-sided flycatchers (*Contopus cooperi*). We used data on breeding phenologies and the composition of insect communities around nests in three regions of Alaska, Anchorage, Fairbanks, and Tetlin National Wildlife Refuge, to compare

C. cooperi breeding stage timing to the emergence of adult aquatic insects. These insects are a potentially important prey resource that may provide highly unsaturated fatty acids (HUFAs), which are hypothesized to be essential for chick development in other aerial insectivore species and may also serve as migration fuel. We used generalized additive mixed models (GAMMs) to build aquatic insect emergence schedules representative of these regions within Alaska. These schedules were compared to maximum observed regional breeding stage windows for *C. cooperi* to evaluate the magnitude of phenological mismatch. Distinct breeding phenologies were exhibited by *C. cooperi* in each region, although mean timing for each breeding stage was not significantly different, indicating most individuals across these regions breed on a similar timeline. Emergence schedules of aquatic insects in Anchorage and Fairbanks were found to shift earlier under forecast warming scenarios for 2033 and 2043. Peak insect emergence in Anchorage occurring during laying, and its projected shift earlier into the laying stage with warming, reduces the availability of aquatic insects later into the breeding season. This limited availability may be mitigated by the high magnitude of daily emergence from streams in Anchorage. *C. cooperi* in Fairbanks displayed higher phenological alignment with local aquatic insect prey, with peak emergence always estimated to occur during an overlap of fledging and departure, and a smaller peak occurring during an overlap of hatching and fledging. Chick development and migration for *C. cooperi* in Fairbanks may still be impacted by projected declines in emergence late in the season, and the lower magnitudes of both emergence and availability of aquatic prey directly at nests.

Human-Bear Conflict in Juneau, AK

Student Investigator: Binta Wold, MS Wildlife Biology and Conservation

PI: Todd Brinkman, IAB/DBW

Funding Agency: ADF&G



Survey participation sticker. Design credit: Binta Wold.

This project was focused on advancing knowledge on human-bear conflict in Juneau Alaska. Our objectives were to quantify perceptions on: the extent of conflict, factors contributing to conflict, and potential solutions. During spring of 2024, we conducted an online survey that included 5,000 randomly selected Juneau residents.

Approximately 750 completed people completed the survey. Sixty-nine percent of participants agreed that there are problems between people and bears in Juneau. The biggest problem was bears getting into trash (90% reported moderate to very serious problem) and other household attractants, followed by people getting too close bears (68% reported moderate to very serious problem). Survey respondents were more concerned about conflict leading to bears being killed (79%) than concerned about the safety of

people around bears (53%). Most respondents were satisfied (57% satisfied vs. 12% unsatisfied) with how ADF&G manages bear in Juneau. Of 11 different interest groups relevant to human-bear conflict in Juneau, respondents most agreed that individual Juneau residents (84% agreed) and landlords (82% agreed) need to do more to address human-bear conflict in Juneau. Final results will be submitted for publication during 2027. For public dissemination, results will be posted at AKBearSurvey.org.

Ongoing Wildlife Studies

Evaluating Spatial-Capture-Recapture Models for Social Carnivores

Post-Doctoral Researcher: Derek Arnold, PhD

PI: Shawn Crimmins, IAB

Funding Agency: ADF&G

In-Kind Support: ADF&G



Group of wolves at rendezvous site on Prince of Wales Island.
Photo credit: M. Kampnich.

Spatially-explicit-capture-recapture (SECR) models are a powerful tool for estimating population density and for understanding the drivers of variation in density. These tools expand on classical mark-recapture methods by incorporating information about animal movements and space-use into the estimation process, providing more precise and less-biased estimates of density. However, violations of sampling assumptions can lead to erroneous conclusions regarding

population density and its drivers. The results of this project have direct applicability to wildlife monitoring not only in Alaska, and not only for wolves, but for a broad range of species and systems. The sampling issues present for wolves in Alaska are not unique, and a robust understanding of how these violations affect modeling results when using SECR has the potential for improving wildlife monitoring globally.

The use of SECR models in Alaska is increasing, with recent applications to wolves in many different game management units. Recent concerns over the accuracy of density estimates derived from SECR models for wolves in Alaska have led to a concern that sampling violations may be occurring when using hair boards to sample these socially structured and highly mobile animals. Using a combination of simulations and empirical data from multiple study sites throughout Alaska, this project seeks to develop new statistical models to determine the degree to which social structure (i.e. subordinate individuals not being sampled) and unique movement behaviors influence SECR estimates.

Models developed for this project are able to accurately identify behavioral/demographic states of animals from spatial detection data and provide a means of generating precise and unbiased density estimates even when presented with behaviorally induced variation in space-use and detection probabilities.

Developing a Multi-Species Monitoring Framework in Southeast Alaska

Student Investigator: Sarah Trujillo, PhD Biological Sciences

PI: Shawn Crimmins, IAB

Funding Agency: ADF&G (RSA S298086, RSA S29254)

In-Kind Support: ADF&G



Wolves captured on a remote camera on Prince of Wales Island.

Photo credit: G. Roffler, ADF&G.

Effective monitoring of large mammals in southeast Alaska is challenging due to accessibility and heavy forest cover. Typical monitoring approaches for wolves, black bear, and deer are largely ineffective in this region, making effective management actions challenging. Moreover, recent listing petitions for wolves in this region necessitate the development of an effective approach for identifying population status. Appropriate

management of harvested species necessitates a robust understanding of population status and change through time. In the absence of such information, management programs are hindered in their ability to appropriately allocate harvest or to assess the vulnerability of threatened populations. Thus, a robust monitoring framework for large mammals in southeast Alaska is critical not only for harvest management, but also for status assessments of potentially threatened species. The objective of this study is to develop an integrated monitoring framework for large mammals on Prince of Wales Island, with an emphasis on wolves, that can be applied throughout southeast Alaska. We are using a combination of motion-capture and time-lapse images to develop density models for wolves, black bear, and black-tailed deer using a suite of analytical approaches including recently developed space-to-event and time-to-event models. We were able to deploy 21 cameras throughout Prince of Wales Island during our pilot field season in fall of 2021, 50 cameras in summer of 2022, 83 cameras in summer of 2023, and 100 cameras in summers of 2024 and 2025. All photos from 2021–2024 have been tagged and processed with processing of 2025 data ongoing. The results of this project will provide the basis for effective long-term monitoring of ecologically, economically, and culturally important large mammals in southeast Alaska. Additionally, this project will provide information on the ecological relationships between predator and prey species in southeast Alaska.

Effects of Human Visitation on Bear Activity at Anan Observatory

Student Investigator: Kayleigh McCarthy, MS Wildlife Biology and Conservation

PI: Shawn Crimmins, IAB

Funding Agency: USDA Forest Service

In-Kind Support: USDA Forest Service, DBW



Black bear feeding on salmon at the lower falls of Anan Observatory. Photo credit: S. Crimmins.

The impacts of recreational activity on wildlife can be substantial. In areas with high concentrations of humans and wildlife, there may be high potential for human activity to negatively affect wildlife. At the Anan Observatory, near Wrangell, Alaska, the USDA Forest Service recently renovated their viewing platform at a substantial cost to the ranger district. The Anan Observatory is a highly regulated bear viewing site in the Tongass National Forest, with visitation controlled through a permit system, however the construction of a larger viewing platform has led

to calls for increasing the number of permits, and thus the human visitation rates. In the summers of 2023 and 2024 we used remote cameras deployed at the observatory and at the upper falls, an area off limits to recreational visitation approximately 0.5-km upstream, to quantify bear activity patterns during the tourist season. Preliminary generalized linear models of bear occurrence indicates that black bear activity is suppressed by the immediate presence of brown bears. Additional analyses of these data are in progress. We are also using long-term monitoring data from the observatory to determine if chronic human visitation has affected bear feeding activity. Preliminary results from this analysis suggest a positive relationship between human visitors and bears at the observatory, potentially indicating a human response to bear presence. Additional analyses will incorporate data regarding salmon abundance to determine if resource fluctuations mediate these relationships. The results of this study will provide the USDA Forest Service with important information that they can use to determine if increased permitting will have deleterious effects on bears at Anan Observatory. There is substantial financial pressure surrounding the management of the observatory, with a number of guiding services operating out of the town of Wrangell, and their financial security is contingent on a long-term management plan that maximizes the number of visitors without having any adverse effects on wildlife.

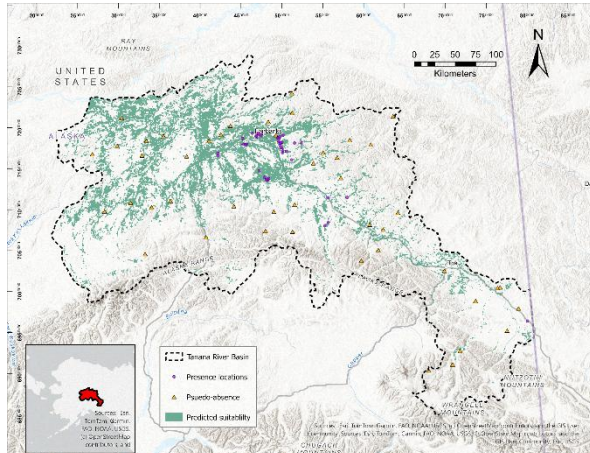
Little Brown Bat Habitat Association Modeling and Geodatabase Development for Adaptive Management

Investigators: Kara White, PhD, Matt Kynoch, MS

PI: Shawn Crimmins, IAB

Funding Agency: USFWS

In-Kind Support: US Army Garrison Alaska – Fort Wainwright



Predicted suitable habitat for little brown bat (*Myotis lucifugus*) in the Tanana River Valley.

The U.S. Fish and Wildlife Service's Fairbanks Field Office, Ecological Services, proposes to assist the US Army Garrison Alaska (USAG Alaska) in refining their fish and wildlife database by identifying GAPS and collating data regarding fish, wildlife, and the habits they occupy. Accurately describing species' association with their habitat is integral to managing impacts on military lands. The USAG Alaska has invested resources to obtain land classification and habitat information, as well as bird and bat habitat use and procuring high quality imagery and LiDAR data to assess changes to habitat over time. Most studies are

over 20 years old, and as a proactive measure, USAG Alaska requires habitat and association information updates and additions to revise current remediation and restoration methods to minimize impacts (Adaptive Management).

The proposed project will collate existing data and produce a geodatabase critical for evaluating future projects and to build little brown bat (*Myotis lucifugus*) habitat models in order to identify additional areas important to this species, which is currently under review for listing on the Endangered Species Act (ESA). The information will lead to increased mitigation success while adapting to a changing climate. Information will include ground truthing previous habitat assessments and reviewing current available LiDAR and imagery data. Information collation, geodatabase development, and map production will then assist in revising standard operating procedures (SOPs) and developing impact avoidance and minimization measures.

Existing monitoring data have been formatted and uploaded to both North American Bat Monitoring Program and Avian Knowledge Network. Two manuscripts resulting from this project have been submitted for peer-review, one focused on the distribution of little brown bats in the Tanana River Valley and another focused on seasonal variation in detection rates and occupancy patterns.

Small Mammal Population Dynamics and Genetic Variation Prior to Development in Alaska National Parks

Student Investigator: Sarah Swanson, PhD Biological Sciences

PI: Shawn Crimmins, IAB and Greg Breed, IAB/DBW

Funding Agency: USGS (RWO 261)

In-Kind Support: NPS, DBW



Small mammal captured in Denali National Park. Photo credit: Mel Flamme (NPS).

Small mammals play a vital role in boreal and Arctic ecosystems in Alaska. As local and resident, keystone species in food webs with quick generation times and small home ranges, small mammals are sensitive bioindicators of local disturbance, including climate change and development. They live year-round in soils and beneath snow, reproduce prolifically and thus, may reveal responses to local environmental changes more quickly than larger fauna. As primary consumers, they influence many trophic levels, including insects, mushrooms, plants, seeds, and bark. Small-mammal biota are vital prey for over-wintering species (raptors, meso-carnivores, carnivores). Despite their importance in the Arctic and boreal food webs, vital ecological data on small mammal populations in Alaska remains scant.

In collaboration with the Nebraska Cooperative Fish and Wildlife Research Unit, we will use advanced genetic approaches to quantify genetic diversity indices, genetic structure, spatial genetic structure, and effective population size across time using archived samples collected as part of Denali National Park's long-term small mammal monitoring program. We will identify individuals with mixed ancestry (if present) and use simulated assignment probabilities for first generation hybrids and backcrosses to better determine whether interspecific admixture assignments are due to hybridization or retained ancestral variation.

Preliminary analyses indicated that voles display large-scale genetic isolation by distance but that these differences are minimized within habitats. Preliminary analyses of genetic richness as a function of population boom-bust cycles suggest that inbreeding coefficients may track population cycles but that allelic richness at the population level remains fairly static over time. Future analyses will focus on hybridization and identifying selective pressures over time.

Distribution and Habitat Use of Nesting Surfbirds in the Steese National Conservation Area

Student Investigator: Sam Simon, MS Wildlife Biology and Conservation

PIs: Shawn Crimmins, IAB and Greg Breed, IAB/DBW

Funding Agency: BLM (CESU Agreement & RWO 264)

In-Kind Support: BLM, USFWS



Viewing a surfbird (*Aphriza virgata*) in the Steese NCA.

Credit: Sam Simon.

As part of the BLM's Strategic Plan for Migratory Bird Conservation, an understanding of the distribution and status of alpine nesting shorebirds within the Steese National Conservation Area (NCA) and White Mountains National Recreation Area have been identified as critical management needs for the Eastern Interior Field office. Very little is known of the distribution or nesting habitat of alpine nesting shorebirds in the Steese NCA, or more broadly in interior Alaska.

This project will entail surveys of upland and alpine habitats focused within the

Steese NCA (where access can be obtained from trails and nearby roads) to document presence and absence of alpine shorebirds and the distribution of territories and nesting locations. We will also capture surfbirds during nesting and brood-rearing to collect breeding ecology and demographic data, and will affix satellite transmitters to a subset of these captured individuals to quantify habitat use, movements and migratory pathways for these individuals. We will develop habitat models using data from satellite tagged individuals to determine summer nesting and brood rearing habitat associations. Habitat use information will be used to quantify the probability of use of surfbirds across the Steese and White Mountains. Using data from surfbirds captured in the summers of 2024 and 2025, preliminary habitat models indicate that nesting surfbirds are exclusively located in high-elevation alpine habitats associated with rocky outcroppings and that, within these areas, habitat use is largely driven by vegetative cover. The information gained from this project will help BLM achieve their management and conservation objectives while also filling the identified role of National Landscape Conservation System lands as outdoor laboratories. This information will also supplement data being gathered on these alpine shorebirds during migration and wintering, allowing a better assessment of the limitations and importance of the Steese NCA in maintaining stable population sizes.

Space-Use of Muskox in Northwest Alaska¹

Student Investigator: Christie Osburn, MS Wildlife Biology and Conservation

PIs: Shawn Crimmins, IAB and Megan Milligan

Funding Agency: ADF&G (Wildlife Base)

In-Kind Support: ADF&G



Resting muskox near Seward, AK. Photo credit: Shawn Crimmins.

Muskox in northwest Alaska are the product of reintroduction efforts initiated in the 1970s for the purpose of returning muskoxen to their native landscape and restoring a source of terrestrial game for subsistence and recreational users.

Fifty years post-introduction, the abundance and geographic range of the population has expanded but many questions remain regarding movement patterns and space-use

of muskox, with specific interest regarding the extent of suitable habitat and home-range requirements. The objective of this study is to model the habitat selection, home range, and movement patterns of adult muskox in the Cape Thompson population in northwestern Alaska. Data collection began in summer of 2026 with the capture and collaring of 30 cow muskox. Collars will collect 4 locations per day, facilitating seasonal comparisons of space-use patterns. The information gained in this project will provide some of the first quantitative data regarding muskox space-use patterns in northwest Alaska and will help identify and delineate suitable muskox habitat and space-use requirements.

¹ Initial project focused on Dall sheep space-use but shifted to muskox in 2026 due to logistical constraints

Synthesizing Ungulate Movement and Migration Across Alaska

Student Investigator: Miles Spathelf, PhD Biological Sciences

Advisors: Shawn Crimmins, IAB and Megan Milligan

Funding Agency: ADF&G

In-Kind Support: ADF&G



Caribou migrating during early spring. Photo credit: S. Crimmins.

Ungulates, especially caribou, in Alaska have some of the longest terrestrial migrations in the world. Human perturbations from roads, mining efforts, hunting, and oil development all have an influence on these vast movements on the landscape undergoing climate change. GPS telemetry data has been collected for many years on numerous herds and has yet to be analyzed over the long time scale ADF&G's data covers. Both movement corridor and barrier behavior analyses can shed light on impacts ungulates are facing

when moving between areas critical for their life histories. Appropriate management of harvested species requires understanding of where these creatures migrate and what barriers they encounter. Vast stores of energy are used for these movement behaviors and can drastically impact survival and reproduction. The objective of this study is to analyze centuries of caribou-years of GPS telemetry data for changes in migration corridors and barrier behaviors as they have arisen on the landscape. Telemetry data from multiple studies and biologists will be compiled and organized for movement analyses and behavior analyses providing both analytical output as well as a dataset for future studies to be conducted more readily. We have obtained, organized, and cleaned decades of caribou herd location information and are conducting movement corridor analyses and will be completing adapted barrier behavior analyses. Preliminary analyses have identified migration corridors and "pinch-points" for caribou in southcentral Alaska. Future analyses will focus on the identification of migration barriers and how these perceived barriers alter movement and migratory behavior. The results of this project will provide corridor information to inform future infrastructure work to help reduce impacts on highly variable herds that hunters rely on for subsistence or are supplemental to hunter's food security. The results may also be useful in comparing ungulate crossing strategies that lessen impact on ungulates.

Identifying Stopover Habitat for Lesser Yellowlegs

Student Investigator: Hannah Vincelette, MS Wildlife Biology and Conservation

Advisors: Shawn Crimmins, IAB and Megan Milligan

Funding Agency: USFWS

In-Kind Support: USFWS



Banded Lesser Yellowlegs. Photo credit: Zac Pohlen.

Lesser Yellowlegs are long-distance neotropical migrants experiencing an estimated 5.3–9.2% annual decline with >60% loss in abundance since 1980. Lesser Yellowlegs are now the focus of the International Lesser Yellowlegs Working Group, which aims to evaluate and address threats to Lesser Yellowlegs population growth, such as habitat conversion, agrochemical application, and climate change. GPS tracking data and citizen science data have revealed the Prairie Pothole Region (PPR) of North

America as an important stopover site, with up to 45% of the Lesser Yellowlegs population passing through the PPR during the peak post-breeding migratory season. The PPR is a vast area of depression wetlands that has been heavily transformed by agricultural expansion and consequential wetland drainage/infilling. However, conservation programs implemented in the PPR have demonstrated the potential of agricultural management practices to restore wetland functions. As climate change is expected to reduce wetlands in the PPR further, agricultural wetlands have the potential to be a powerful conservation tool for declining species such as Lesser Yellowlegs.

We will use GPS tracking and citizen science data to 1) Determine critical habitat features for Lesser Yellowlegs in the PPR, and 2) understand the impacts of future land use and climate change on Lesser Yellowlegs' distribution in the PPR. Preliminary analyses suggest that stopover habitat selection is a scale-dependent process for Lesser Yellowlegs and that a combination of landcover and open water factors are strong drivers of stopover habitat. This work will contribute to the Lesser Yellowlegs Strategic Recovery Plan currently in development and provide critical information for landscape conservation planning and prioritization in the Prairie Pothole Region.

Habitat and Density Modeling of Brown Bears

Post-Doc: Derek Arnold, PhD

PI: Shawn Crimmins, IAB

Funding Agency: ADF&G (RSA 1126076)

In-Kind Support: Data provided by ADF&G



Brown bear (*Ursus arctos*) in interior Alaska. Credit: Shawn Crimmins.

Estimating density of wide-ranging species such as bears is challenging, particularly in remote landscapes with limited access. From 2019-2023 ADF&G completed two large-scale brown bear research projects in southcentral Alaska. The first entailed fitting adult bears with radio-collars primarily for use in a 2022 mark-resight survey, the second entailed non-invasive genetic sampling during the summers of 2020-2022. We hope to use these two datasets to develop models of brown bear density and habitat use that could

help guide harvest management strategies.

Habitat models will be developed using a combination of resource selection functions and integrated step-selection-analyses with a suite of both static and temporally varying spatial covariates including vegetation structure, topography, hunting risk, and climatic variables such as snow cover. Habitat models will include random effects for individual animal and year to account for unequal sample sizes. Density estimation will be conducted using spatially-explicit-capture-recapture models with a suite of static and temporally varying predictors for both the detection and density processes. Model selection for both habitat and survival models will be conducted using an information theoretic framework.

Preliminary analyses indicate that bear density is largely unaffected by many habitat covariates, likely reflecting the generalist nature of the species but perhaps reflecting sparse data that is insufficient to capture variability in bear habitat. Initial habitat use models from telemetry data suggest that bears respond to some habitat covariates and that the inclusion of this information in spatial-capture-recapture models improves model precision.

Completed Ecological Studies

Arctic Biogeochemical Response to Permafrost Thaw (ABRUPT)

Research Technician: Colin Edgar, IAB

PI: Eugénie Euskirchen, IAB

Funding Agency: USGS (RWO 251), NSF LTREB, NSF LTER



The Bonanza Creek Fire approaching the field sites during June 2025. Image from the webcam installed at the internal collapse scar bog site.

Our objective is the continued collection of eddy covariance and associated meteorological and biophysical information across four sites spanning a gradient of permafrost regime to better understand the controls over carbon, water, and energy cycling in a lowland boreal landscape. Our sites include a black spruce forest with permafrost, two internal collapse scar bogs of different ages formed following thermokarst, and a rich fen without near-surface permafrost. These data have been collected beginning in 2010.

The Bonanza Creek Fire began on June 21, 2025 (shown above), with the fire coming within about one kilometer from our four sites. Access to the sites was restricted until mid-August 2025. Fortunately, our sites were not impacted due in large part to close coordination between Bonanza Creek LTER personnel and fire managers and the establishment of a containment line. Our instruments continued to run successfully, with sustained measurements of the carbon, water, and energy fluxes.

Our data have been made available through the Ameriflux website (<https://ameriflux.lbl.gov/>), including recent measurements collected through 2024. We are currently processing data from January 2025 – August 2025 in an effort to make data available for a 2025 synthesis of data across flux tower sites around the Arctic. In 2024, our data were included in the Arctic Report Card (<https://arctic.noaa.gov/report-card/report-card-2024/arctic-terrestrial-carbon-cycling/>), which showed that the Arctic has become a net source of carbon due to both wildfires and emissions of methane. These findings are also part of another publication that is currently in review at *Environmental Research Letters*. Furthermore, our flux data are included in a recent study of sublimation of tundra and taiga snowpack as well as a pan-Arctic synthesis that shows the decreasing sink strength of high-latitude terrestrial regions. Overall, our set of long-term, year-round measurements along this gradient of permafrost thaw are key in understanding changes in terrestrial carbon cycling in a vulnerable region.

List of Abbreviations

ACRC	Alaska Coastal Rainforest Center
ADF&G	Alaska Department of Fish and Game
AFS	American Fisheries Society
AKCFWRU	Alaska Cooperative Fish and Wildlife Research Unit, USGS
BIA	U.S. Bureau of Indian Affairs
BLaST	Biomedical Learning and Student Training, UAF
BLM	U.S. Bureau of Land Management
CASC	Climate Adaptation Science Center, USGS
CESU	Cooperative Ecosystem Studies Unit, UAF
CFOS	College of Fisheries and Ocean Sciences, UAF
CRITFC	Columbia River Inter-Tribal Fish Commission
DBW	Department of Biology and Wildlife, UAF
eDNA	environmental DNA
EPSCoR	Established Program to Stimulate Competitive Research, NSF
GPS	Global positioning system
GRFP	Graduate Research Fellowship Program, NSF
IAB	Institute of Arctic Biology, UAF
IALT	Interior Alaska Land Trust
INBRE	IDeA Networks of Biomedical Research Excellence, NIH
LTER	Long-Term Ecological Research Network, NSF
LTREB	Long Term Research in Environmental Biology, NSF
NCA	National Conservation Area
NIWR	National Institutes for Water Resources
NPS	U.S. National Park Service
NSF	National Science Foundation
OSM	Office of Subsistence Management, USFWS
PI	Principal Investigator
PIT	passive integrated transponder
PNWRS	Pacific Northwest Research Station, USFS
POW	Prince of Wales
PPR	Prairie Pothole Region
PSMFC	Pacific States Marine Fisheries Commission,
R&M	Research and Monitoring program, USFWS
RSA	Reimbursable Services Agreement
RTS	Research and Technical Support, Division of Sport Fish, ADF&G
RWO	Research Work Order
SECR	Spatially-explicit capture-recapture
SSP	Science Support Program, USGS and USFWS
TCC	Tanana Chiefs Conference
TVWA	Tanana Valley Watershed Association
UAF	University of Alaska Fairbanks
UAS	University of Alaska Southeast

URSA	Undergraduate Research and Scholarly Activity program, UAF
USAG	U.S. Army Garrison
USDA	U.S. Department of Agriculture
USFS	U.S. Forest Service, USDA
USFWS	U.S. Fish and Wildlife Service
USGS	U.S. Geological Survey
VetMed	Veterinary Medicine Program, UAF
WERC	Water and Environmental Research Center, UAF